

7/7 ch-1
APV

* DBMS:

→ Collection of interrelated data, and a set of programs to access those data.

* Goal of DBMS:-

→ Convenience and Efficiency.

* Purpose of DBMS:-

→ Earlier → File system was being used.

① Redundancy (Duplicate) :-

→ Data can be ~~repeated~~ repeated.

→ Disadvantage: Inconsistency. (When having different record of same entity).

② Difficult access:

→ File statements are difficult to access.



③ Data isolation problem:

→ Data is integrated together and they are not isolated.

④ Integrity problem:

→ set some rules on something → Integrity

→ Some constraints can be forgotten.

⑤ Atomicity problem:

→ two bank account → GP to Friend → 100\$ paying

→ We want the entire process to happen 100% or it should not happen at all.

⑥ (Parallel) Concurrent Access Problem:

→ two transaction happens at the same time, then it can happen.

⑦ Security problem:

→ Generally files are open for everyone.

* Views of data:

① Physical view / level:

→ It describes how the data is stored.

② Logical view / level: data

→ It describes what is stored (columns, rows, ...)

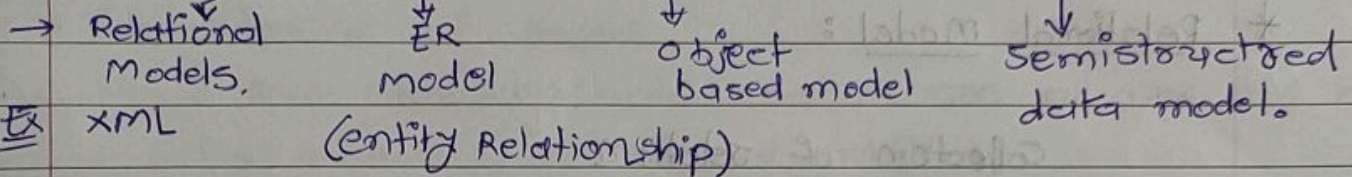
* Physical data independence:

That means physical level is totally hidden from logical level. (If physical level changes it does not affect logical level.)

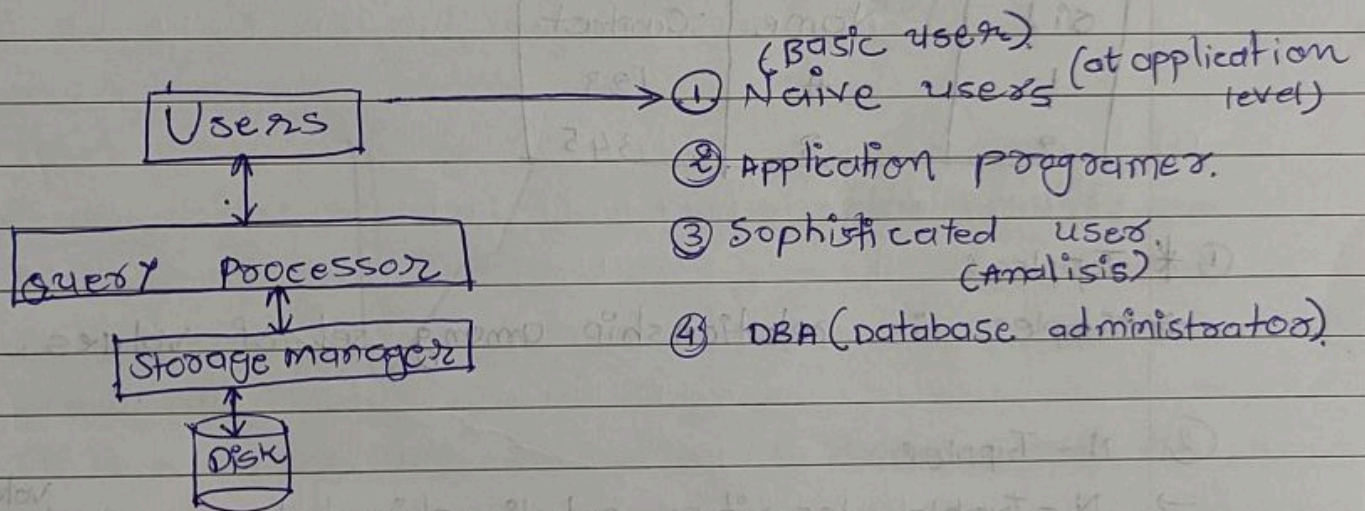
③ View level:-

→ Creating different views of your database in a number of times.

* Data Models :-



* Database architecture.



① Schema definition: Basic skeleton like structure of database.

② storage structure definition: In what manner you want to store data.

③ Schema and physical structure modification: changing/modifying database on the need of users.

④ Back Granting Authorization: DBA will define a set of roll for different person to access different part of data.

⑤ Routine maintainers:

→ Take care of the server.

ch-2* Relational Model:

Collection of relation.

Attributes
 tuples
 properties

Sid	Name	Contact
1	A	123
2	B	345

① * Tuples:

→ Tuple is relationship among set of values.

② N-Tuples:

→ N-Tuple describes relationship between n-values.

③ Relation Instance:→ An instance of Relation is called Relation Instance.
Specific④ Domain:

→ For each attributes, there is a set of permitted values that is called domain.

⑤ Atomic Domain: (when it can not be divided).

→ Domain is called atomic if elements of the domain are indivisible. (Ex id, salary, marks)

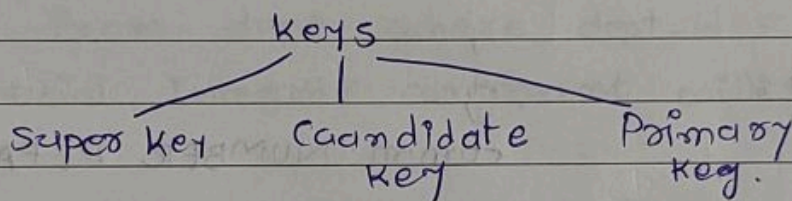
⑥ NULL values: ① Not applicable ② Not known

→ NULL is a special value which signifies that values is unknown or does not exist.

⑦ Keys

→ Importance of keys:

→ ^{No} ~~know~~ two tuples in the relation are allowed to have exactly same value for all attributes.



(i) Super key:

→ It is set of one or more attributes which identify a tuple uniquely.

(ii) Candidate key:

→ It is a super key for which no proper subset is a super key.

(iii) Primary key:

→ candidate key selected by DB designer is called as primary key.

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Data base designing :

① Initial phase :

→ user requirement.

② Conceptual Design $\Rightarrow$ E-R model

③ Specification of functional requirement

→ Convert user ∇

④ logical designing

→ Relational data model.

⑤ Physical design

EMP-ID Name Dept Building Budget

* E-R model:

- ① Entity Set → set of entities of same type. and they
- ② Relationship set share same attributes
- ③ Attributes
- ④ Constraints

* Entity: Any object or thing in real world.

② Relationship-set: Person owns car.

→ Relationship set is a set of relationship of same type.

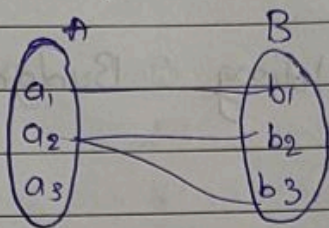
- ③ Attribute
- Simple (marks, grade, dept-name)
 - Composite (can be divided further)
 - Single valued attribute
 - Multi valued attribute. (Phone-no, Email-ID)
 - Derived attribute (DOB - Age)
 - Stored Attribute (DOB).

- ④ Constraints
- mapping cardinalities
 - one to one
 - one to many
 - many to one
 - many to many
 - Participation.

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* Cardinality Participation:

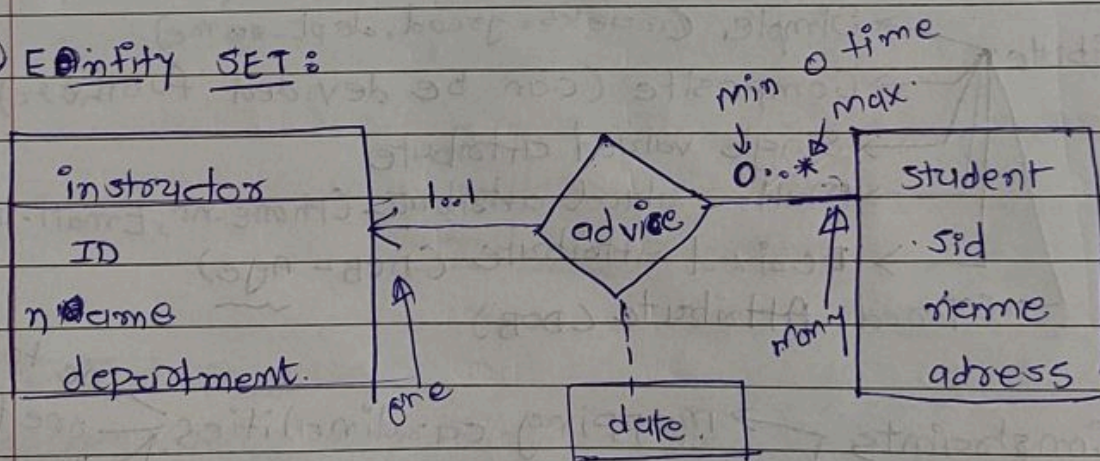
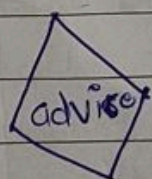
Total
 Partial



→ An entity set E in relationship R (Relationship set) is said to have total participation if every entity of E participates with at least one relationship.

* Partial participation

→ If some of the entities are participating then it is said to be partial participation

* E - R Notation① Entity SET:② Relationship set:-

instructor advise student

* $\rightarrow$ Instructor may advice at ~~the~~ most one student
 $\rightarrow$ Student may have many advisers.
 Many to one

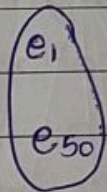
* Instructor can advice many students.
 $\rightarrow$ student may have atmost one adviser.
 One to many

* Instructor can advice many students.
 $\rightarrow$ students ^{can} ~~may~~ have many advisers.
 many to many.

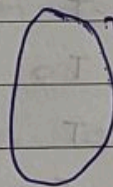
$\Rightarrow$ $0..* \rightarrow 1..1$ (one to many)

* weak entity sets:-

employee



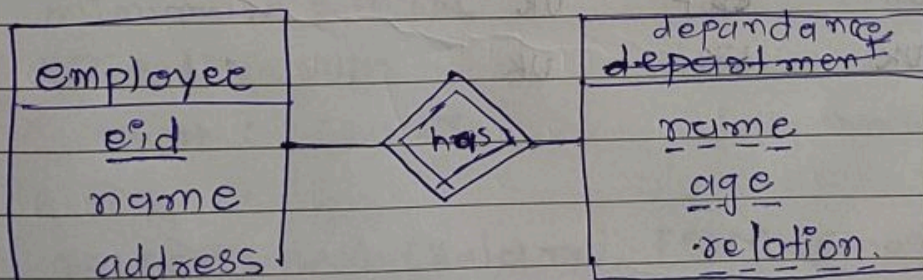
dependance



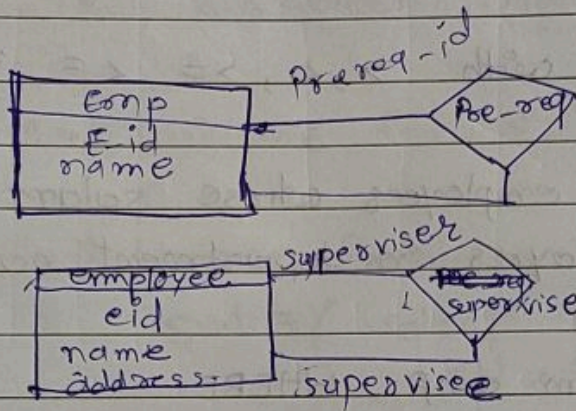
name, age, relation

a 20 son

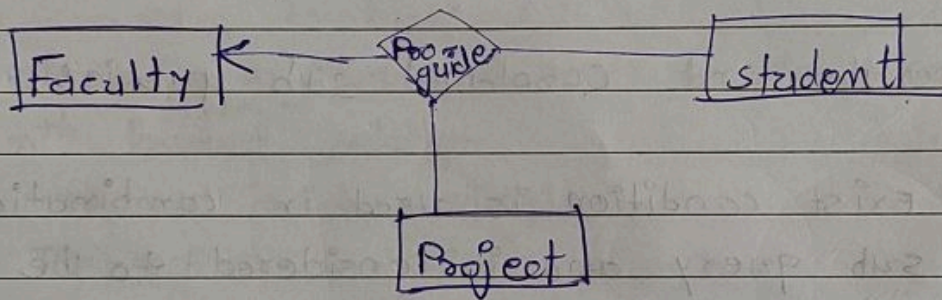
$\rightarrow$ when there is no uniqueness.



* Weak entity set



* Non-binary relationship set (n-ary)



* Reduction of Relation to Relational schema

- n entity set
 - m relationship set.
- } n+m

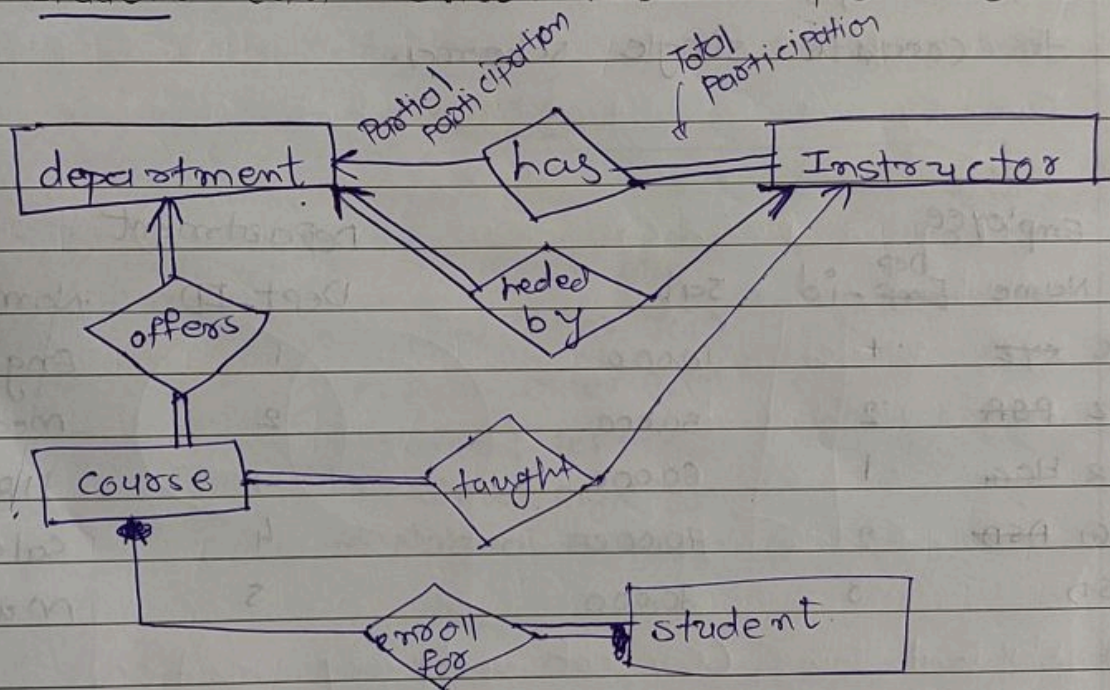
Student
ID
name
DOB
{ Phone }
no
Age

deivable

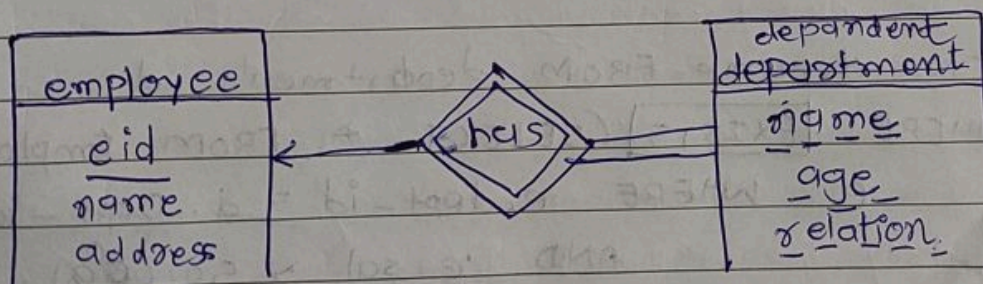
student (ID, Pname, mname, lname, date of birth)
 ↓
 Primary Key.

* Create E-R diagram for university management system

- It has many departments. each department has multiple instructors. one among them is head of the department.
- An instructor belongs to only one department.
- each department offers multiple courses. each of them is taught by a single instructor.
- A student can enroll for multiple courses.

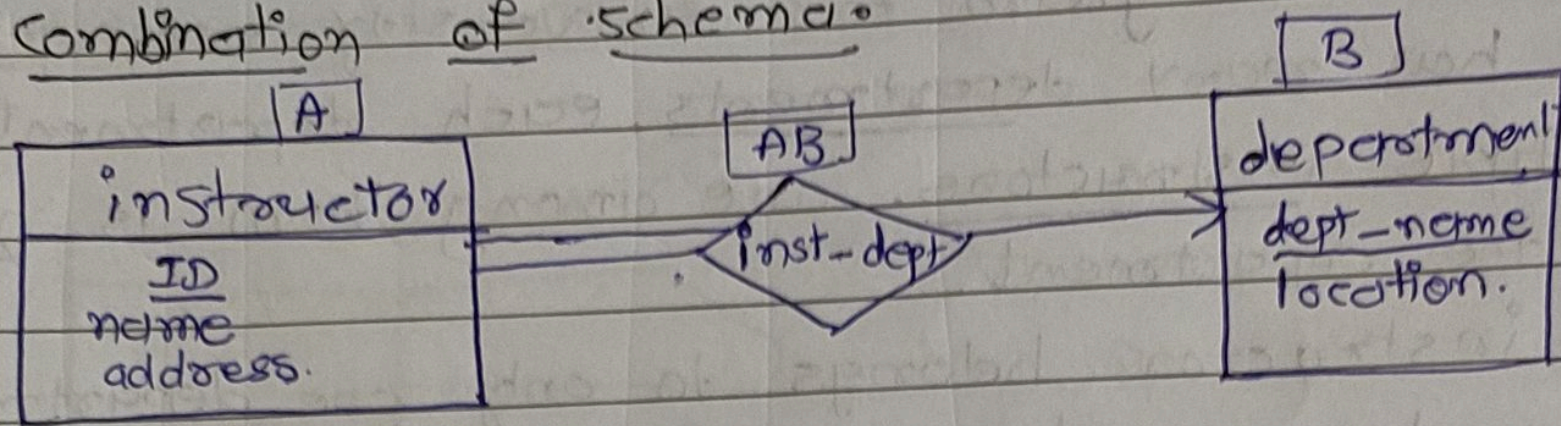


* Redundancy of schemas:





* Combination of schema:



- (1) If A has total participation in many to one relationship, then it can be combined with AB to create single schema.

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APV

→ For one to one relationship set the attribute is merged with either of the entity set.

→ For many to one or one to many we merge the attribute with the entity on many side.

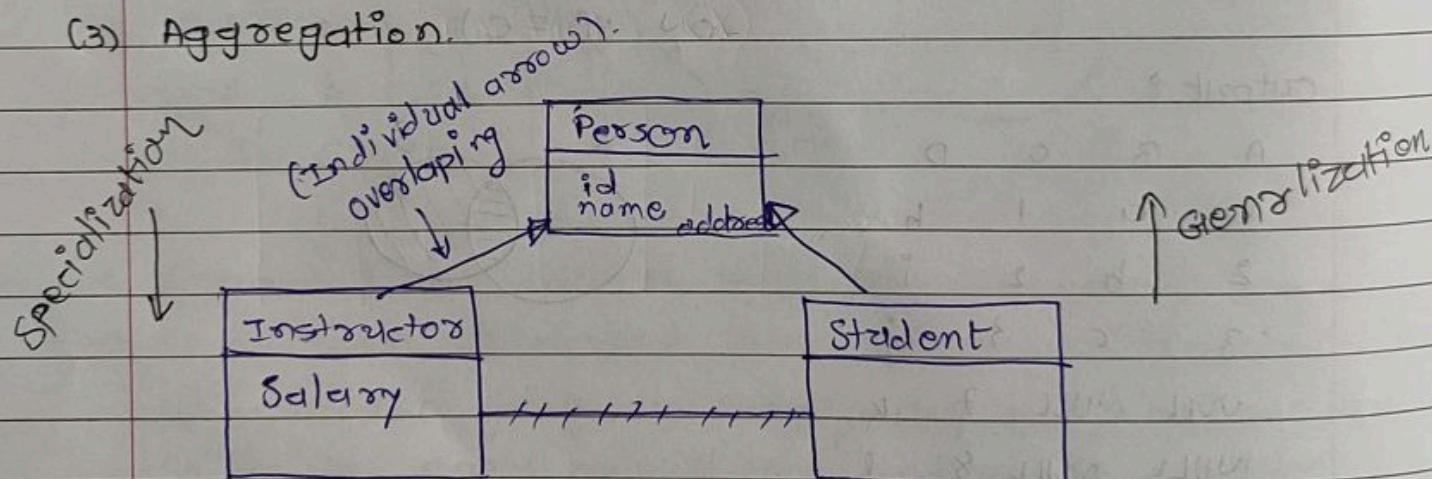
→ In case of many to many we keep the attribute within relationship set only.

* ch-6 Extended E-R Features.

(1) Specialization

(2) Generalization

(3) Aggregation



* Specialization:

→ It is process of defining sub groupings within single entity set.

* overlapping specialization:

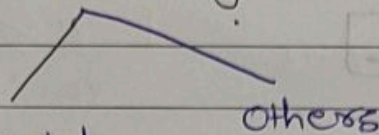
→ In overlapping specialization parent entity can belong to more than one child entity.

$\wedge \rightarrow \text{and}$
 $\vee \rightarrow \text{OR}$
 $\neg \rightarrow \text{NOT}$

* Parent Disjoint:

→ when parent entity can belong to only one child entity.

* ch=6 · Relational Algebra



- | | |
|---|--|
| <ul style="list-style-type: none"> - select - project - Union - set difference - Cartesian product - rename | <ul style="list-style-type: none"> - Intersection - Natural Join - assignment |
|---|--|

(1) Select: (σ (lower case sigma)) (where of sql)

→ select operation selects the tuples which satisfy given predicate. (similar to where in sql)

→ " $\sigma_{\text{predicate}}(\text{relation})$ "

→ Ex $\sigma_{d_name = 'CE'}(\text{instructor})$

→ select all instructor with salary > 90,000

$\sigma_{\text{salary} > 90,000}(\text{instructor})$

$\sigma_{d_name = 'CE'}(\sigma_{\text{salary} > 90,000}(\text{instructor}))$

(2) Project: ($\pi_{a_1, a_2, \dots, a_n}(\text{relation})$) (select of sql)

→ This operation returns its argument relation with certain attributes left out.

→ Find the names of all instructors in ce department.

$\pi_{\text{name}}(\sigma_{d_name = 'CE'}(\text{instructor}))$

→ Find all courses taught in 2022 odd sem, 2022 even sem OR Both.

③ Union

$\boxed{\text{exp1}} \cup \boxed{\text{exp2}}$

→ We should check compatibility in $\cup, \cap$, set difference.
must have same no. of columns in both tables.

29/x ④ Set difference:

→ Set difference finds such tuples which are present in one relation but not in other.

→ Find all courses taught in 2022 odd sem but not in 2022 even sem.
~~must have~~

⑤ Cartesian product: (X)

emp

A	B	C	D	E
1	a ₁	c ₁	1	x
2	a ₂	c ₂	2	y

A	B	C	D	E
1	a ₁	c ₁	1	x
1	a ₁	c ₁	2	y
2	a ₂	c ₂	1	x
2	a ₂	c ₂	2	y

spurious tuples.

$\sigma_{A=D}(\text{emp} \times \text{dep})$ (Join operation).

⑥ Set intersection. (derived operation, $A - (A - B) = A \cap B$).

→ Find all courses taught in 2022 odd sem and 2022 even sem.

⑦ Natural Join: ($\bowtie$)

→ natural join first takes ~~to~~ cartesian product of two arguments.

→ Then it performs selection based on such attributes which are present in both relation.

→ Finally ~~but~~ duplicate attributes are removed.

→ When natural join is performed without common attribute, the result will be cartesian product.

⑧ Rename: (ρ)

① $S_x(E)$

② $S_{A_1, A_2, \dots, A_n}(E)$

⑨ Assignment: ($\leftarrow$)

Temp $\leftarrow \sigma(x)$; $\Pi(\text{Temp});$ } Breaking down of query.

Left outer join ($\leftarrow$)

Right outer join ($\rightarrow$)

Full outer join ($\bowtie$)



① Generalized projection:

→ Find out all employees ids, names and monthly salary.

$\pi_{ID, name, salary} (employee)$

$\pi_{fns} (\sigma)$

*

Find avg salary in each dept.

Select avg(salary) FROM employee group by dept name

$G_{(group\ by)}$

$d_name G_{avg(salary)} (employee)$